A-A-53999
10 October 1989

COMMERCIAL ITEM DESCRIPTION

INSERT, HANDPIECE, DENTAL, ULTRASONIC PROPHYLAXIS UNIT

The General Services Administration has authorized the use of this Commercial Item Description.

Salient characteristics:

This commercial item description covers dental handpiece inserts suitable for use as replacements in dental ultrasonic prophylaxis units, (see Notes). The inserts shall be No. 3 Flat-Hoe Chisel type, or No. 1000 Curette type, and shall be suitable for removing calculus and scale from tooth enamel.

Materials. The tip shall be fabricated of class 4, type 420, martensitic stainless steel in accordance with ASTM Designation F899. The connecting body of the No. 3 Flat-Hoe type insert shall be fabricated of Monel 400 (AMS 7233) in accordance with ASTM Designation B127. The connecting body of the No. 1000 Curette type insert shall be fabricated of class 5, type 630, precipitation hardening corrosion resisting steel (AISI 17-4 PH), in accordance with ASTM Designation F899. The nozzle shall be fabricated of polysulfone, and the transducer of Permanickel Alloy 300 strips.

Style and design. The inserts shall consist of a tip, which is either No. 3 Flat-Hoe Chisel type or No. 1000 Curette type as specified, a nozzle, a connecting body, and a transducer (stack). The No. 3 Flat-Hoe Chisel type insert shall be of a design appropriate for removing supragingival calculus from labial and lingual posterior tooth surfaces, as well as surface stain from all accessible tooth surfaces. The No. 1000 Curette type insert shall be of a design appropriate for removing calculus from all supragingival and shallow supragingival tooth surfaces.

AMSC N/A FSC 6520

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The inserts shall include an anti-rotation key, to prevent rotation of the tip. The operative tip shall be in line with the central axis of the handle. The flow through the openings in the collar around the tip, working in conjunction with the metal eyelet at the junction of the tip and nozzle shall assure maximum cavitation of water at the tip with a minimum formation of droplets at the junction of the tip and nozzle.

The tip shall be activated by mechanical oscillations of the strips which make up the transducer. These oscillations result in elliptical tip motions at a frequency of 25,000 tip motions per second over a distance of 0.003 inch. The stack shall be 0.159 to 0.168 inch thick. Each individual strip in the stack shall be 0.0095 to 0.0105 inch thick, and 0.178 to 0.187 inch wide. The transducer shall convert the low direct current voltage output from the prophylaxis unit (see Notes) into the elliptical motions of the insert tip.

Hardness. The operative tip shall have a Rockwell hardness of C40 to C45 when tested in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers Pyramid).

Autoclavability. The complete insert shall show no evidence of corrosion after being cleaned in accordance with manufacturer's instructions and subjected to three cycles of steam autoclaving at 121 degrees (^{o}C), 15 psi for 30 minutes each.

Identification marking. Each insert shall be marked with the company name or trademark.

Finish. The operative tips shall have a mirror or bright finish.

Workmanship. The nozzle shall show no evidence of cracking, holes or pitting. The transducer shall be free from metal slivers, burrs, or other defects which may detract from its appearance or impair its serviceability. The tips shall not have any sharp edges.

Unit. Each (EA). One insert of one style only, as specified, constitutes one unit.

Unit package. Each insert shall be individually blister-packaged in a manner which precludes damage during shipping, handling and storage.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, at inspection level II. The unit of product for examination purposes shall be one handpiece insert of each style. Each insert shall be examined for defects including, but not limited to, those listed below.

Defects.

Identification marking missing, incomplete, or incorrect.
Insert does not fit properly into handpiece as specified.
Workmanship not as specified.
Anti-rotation key missing or does not prevent tip rotation.
Tip motions not elliptical.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, at inspection level S-1. The unit of product for test purposes shall be one handpiece insert of each type.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations.

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NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-01-043-5022	INSERT, HANDPIECE, DENTAL ULTRASONIC PHOPHYLAXIS UNIT, No.3, Flat-Hoe Chisel Type
6520-01-137-8471	INSERT, HANDPIECE, DENTAL ULTRASONIC PROPHYLAXIS UNIT, No. 1000, Curette Type

NOTE: The following National Stock Numbers are referenced by this document.

National Stock Number	Item Identification
6520-01-141-7467	ULTRASONIC PROPHYLAXIS UNIT, DENTAL
6520-01-142-8228	ULTRASONIC PROPHYLAXIS UNIT, DENTAL

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2315

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53999	INSERT, HANDPIECE, DENTAL, ULTRASONIC PROPHYLAXIS UNIT

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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